

Question	Marking Guidance	Mark	Comments
06.1	1. Increases dissociation of oxygen; 2. For <u>aerobic</u> respiration at the tissues/muscles/cells OR Anaerobic respiration delayed at the tissues/muscles/cells OR Less lactate at the tissues/muscles/cells;	2	1. Accept unloading/ release/reduced affinity for dissociation
06.2	1. (Time) 10 minutes; 2. (Ratio) 1.6875(:1); Allow 1 mark for correct ratio calculated from wrong time	2	For the ratio accept any correct rounding
06.3	1. Increase in breathing (rate); 2. Similar/same pCO ₂ per breath, but more breaths; OR 3. Increase in tidal volume; 4. Similar/same pCO ₂ per breath, but increased volume per breath;	2	Award mark points 1 and 2 OR 3 and 4 1. Allow more breaths per minute 1. Reject more BPM 3. Accept each breath is deeper
06.4	Second box ticked (Muscle fibres have a limited amount of phosphocreatine.)	1	

06.5	1. More acetylcoenzyme A would enter the Krebs cycle; 2. (So) the Krebs cycle generates (more) reduced coenzymes OR (So more) reduced coenzymes pass their electrons to the electron transfer chain; 3. (So more) ATP would be produced; 4. Athletes could build (slow) muscle (fibres) without exercising; 5. (Having more) slow muscle (fibres) would increase endurance;	4 max	1. 2. and 3. idea for more is required once 2. Accept examples of reduced coenzymes 2. Reject production of reduced NADP^+ or NADPH_2 4. Ignore 'develop (slow) muscle (fibres) at rest' as in stem of question 4. Accept description of not exercising, eg without training 5. Accept descriptions of endurance in terms of delayed onset of anaerobic respiration
06.6	1. (EPO) causes blood to thicken; 2. (The thickened blood) could block the coronary arteries OR (The thickened blood) slows blood flow OR (The thicker blood) could cause clots;	2	1. Accept descriptions of thickening, eg more viscous 2. Reject atheroma/plaque (forms) 2. Accept could cause thrombus/embolus

06.7	1. Some cyclists will gain a bigger advantage/increase OR Cyclists with a haematocrit of 50% would not be able to gain an advantage; 2. There are health risks (associated with) taking EPO;	2	1. Accept use of the data, or suitable calculations, eg some may have an 8% increase, others 0% 1. Some cyclists might naturally have a haematocrit over 50% (and so not be allowed to compete) 2. Accept dangerous side-effects of taking EPO, or examples of health risks
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